Jothi Teaching Standards

Digital Board Teaching Standard

Operational standard for using the digital board to support clear explanation, diagnostic teaching, and visible reasoning

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Purpose of the Document

This document defines the standard for digital board use within the Jothi Tutor Academy. Its purpose is to ensure that tutors use the digital board not merely as a presentation surface, but as an instructional tool that supports explanation, questioning, monitoring of student thinking, and misconception correction.

In online tutoring, the board often functions as the student's main visual access point to the tutor's thinking. Poor board use weakens clarity even when the tutor understands the subject well. Strong board use, by contrast, helps make reasoning visible, slows teaching to a usable pace, and creates better opportunities for diagnosis and correction.

Intended Audience

This document is intended for:

- tutors delivering online lessons within Jothi Learning
- mentors observing clarity of explanation and board use
- academy leadership overseeing teaching quality and tutor readiness

Related JTA Documents

- JF03 — The 4T Teaching Engine
- JTS01 — Lesson Delivery Standard
- JTS05 — Teaching Environment and Technology Standard
- JTD04 — Mentor Observation Standard

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1. Purpose of the Digital Board in Jothi Lessons

The digital board is not an optional accessory. It is part of how teaching happens at Jothi.

It serves several instructional purposes:

- making reasoning visible step by step
- showing the structure of a method or concept
- slowing explanation to an understandable pace
- helping students follow the logic behind the work
- supporting diagnostic questioning and correction

The board should therefore be treated as a teaching surface, not simply a place to display final answers.

2. Relationship to the 4T Teaching Engine

Digital board use should support all four phases of the 4T Teaching Engine.

Teach

The board helps the tutor explain clearly by making each step visible and logically ordered.

Test

The board supports questioning by allowing the tutor to point to specific steps, hide or omit steps deliberately, or ask the student to justify what is written.

Track

The board helps the tutor observe how the student is thinking when the student completes, annotates, or comments on visible working.

Transform

The board supports misconception correction by allowing the tutor to compare correct and incorrect methods, isolate the point of error, and rebuild the reasoning clearly.

For this reason, board use should be active, purposeful, and closely tied to student understanding.

3. Core Principles of Strong Digital Board Use

Strong digital board work should be:

- legible
- logically sequenced
- uncluttered
- appropriately paced
- focused on reasoning rather than decoration

Students should be able to look at the board and follow how the tutor is thinking. If the board is visually confusing, rushed, or structurally unclear, lesson quality drops even when the explanation itself is technically correct.

4. Legibility and Visual Clarity

Tutors must ensure that writing on the board is easy to read.

This means:

- handwriting should be large enough to read comfortably
- symbols and notation should be written clearly
- spacing should separate one step from another
- diagrams, labels, and annotations should be readable
- important parts of the working should stand out through position and structure, not clutter

Legibility is a minimum requirement. If the student cannot read the board properly, explanation and diagnosis both weaken.

5. Step-by-Step Reasoning

Tutors should use the board to show how an answer or method is built, not merely where it ends.

This means:

- writing steps in a clear sequence
- avoiding unexplained jumps

- showing substitutions, transformations, and intermediate reasoning
- revealing the logic behind each move where needed

Students should not be forced to infer missing thought processes from a compressed set of final lines.

6. Pacing of Board Work

Board writing should be paced for learning, not speed.

Tutors should avoid:

- writing too quickly for the student to follow
- speaking far ahead of what is visible on the board
- filling the page before the student has processed the current step

A well-paced board gives the student time to think, notice patterns, and respond to questions.

7. Using the Board to Support Diagnostic Teaching

The board should actively support diagnostic questioning.

Tutors can use it to:

- ask students to complete the next step
- pause at a key line and ask why it works
- present an incorrect method and ask the student to identify the flaw
- compare two approaches and ask which one is valid
- expose exactly where a misconception begins

This matters because the board can turn passive explanation into visible reasoning and evidence of understanding.

8. Using the Board to Monitor Student Thinking

The board also supports the tracking of student thinking.

Tutors should use it to observe:

- whether the student follows the structure of the method
- whether the student can explain what a line means
- whether the student copies mechanically or reasons independently
- whether hesitation appears at predictable points
- whether the same misconception recurs across similar examples

The board becomes especially useful when the tutor can see not only the final answer, but the route by which the student arrived there.

9. Using the Board to Correct Misconceptions

When a student makes an error, the board should help the tutor correct it precisely.

Effective use may include:

- isolating the exact point where the reasoning went wrong
- rewriting the step more clearly
- contrasting the incorrect method with the correct one
- showing why the correction works
- asking the student to explain the corrected logic

Correction is more effective when the student can see the change clearly and understand why it matters.

10. Subject-Specific Application

10.1 Mathematics

In Mathematics, the board should make procedural and conceptual structure visible.

Tutors should:

- show algebraic steps clearly
- avoid collapsing several transformations into one line when the student still needs to see the movement
- use layout to separate methods, cases, or stages of working

- make diagrams, graphs, and geometric constructions readable

10.2 Physics

In Physics, the board should support both concept explanation and disciplined written method.

Tutors should:

- show formulas clearly
- make units visible
- model substitution step by step
- separate known values, formulas, and working logically
- use diagrams where they genuinely improve understanding

10.3 Chemistry

In Chemistry, the board should support symbolic, structural, and conceptual clarity.

Tutors should:

- write equations clearly and accurately
- keep notation readable
- model balancing and structural representation step by step
- use the board to make relationships between particles, equations, quantities, or processes easier to follow

11. Common Board Work Errors

The following mistakes weaken board effectiveness and should be avoided:

- writing too small to read comfortably
- writing too quickly for the student to process
- overcrowding the board with too much information
- skipping key reasoning steps
- using the board only for final answers
- leaving incorrect work unaddressed when it reveals misunderstanding
- treating neatness as more important than instructional clarity

- writing in a way that the tutor understands but the student cannot follow

These errors often create the illusion of efficient teaching while reducing actual understanding.

12. Student Participation Through the Board

The board should support active student involvement where possible.

Tutors should regularly invite students to:

- explain a visible step
- predict the next line
- identify where a mistake occurred
- justify why a method works
- compare two written approaches
- complete part of a worked solution

The aim is to use the board as a shared thinking space rather than a one-way display.

13. Mentor Observation Focus

When reviewing digital board use, mentors should look for evidence that the tutor:

- writes clearly and legibly
- structures explanation step by step
- uses the board to support diagnostic questioning
- makes student reasoning visible
- uses the board to correct misconceptions effectively
- avoids clutter, rushing, and missing logical steps

Board use should be judged not by neatness alone, but by whether it improves teaching clarity and student understanding.

14. Relationship to the Wider JTA System

This document supports the wider JTA framework by translating part of the 4T teaching model into visible lesson practice.

It is closely connected to:

- JTS01, because strong lesson delivery depends on visible explanation
- JTS05, because technology and board use are operationally linked
- JTD04, because mentors need a clear standard against which board use can be observed

The digital board is therefore both a teaching tool and an observation point within tutor development.

15. Conclusion

At Jothi, digital board use is part of teaching quality, not a minor technical preference. Tutors are expected to use the board in a way that makes reasoning visible, supports diagnostic questioning, helps track student thinking, and strengthens misconception correction.

This standard exists to ensure that visual explanation remains clear, purposeful, and instructionally useful across the academy.